

Experiment Comparison Report

This report compares two Deep Learning experiments for MNIST Handwritten Digit Recognition. Experiment 1 used the ReLU activation function with a learning rate of 0.0001, while Experiment 2 used the Tanh activation function with a learning rate of 0.00005.

Feature	Experiment 1	Experiment 2
Learning Rate	0.0001	0.00005
Activation Function	ReLU	Tanh
Epochs	10	10
Optimizer	Adam	Adam
Loss Function	CrossEntropyLoss	CrossEntropyLoss
Final Train Accuracy	96.48%	96.87%
Best Validation Accuracy	97.08%	96.79%
Final Validation Accuracy	96.86%	96.79%
Final Validation Loss	0.1067	0.1030
Final Test Accuracy	97.11%	96.98%
Final Train Loss	0.1101	0.0982
Performance Stability	More Stable	Slight fluctuations
Overall Result	Better Generalization	Better Training Fit

Analysis

Experiment 1 achieved the highest overall performance with better validation and test accuracy. The ReLU activation function helped the model converge faster and provided more stable learning behavior.

Experiment 2 achieved slightly higher training accuracy and lower training loss, but its validation and test accuracy were slightly lower. The smaller learning rate and Tanh activation function resulted in slower convergence.

Conclusion

Experiment 1 demonstrated better generalization performance and more stable validation results, making it the best-performing model overall for the MNIST handwritten digit classification task.